Numpy Library

MSc. Thanh-An Nguyen
BSc. Nhat-Tan Bui

Software Engineering Laboratory (SELab)
University of Science,
Vietnam National University, Ho Chi Minh city

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Introduction

- NumPy (Numerical Python): core library for arithmetic calculations, vector, matrix, etc.
- Commonly used in data processing and analysis
- The foundation of other libraries such as OpenCV, Pandas, ...

Numpy Array

- Import Numpy in Python

```
import numpy as np
```

- Create matrix/vector

```
a = np.array([1, 2, 3], dtype=np.float)
b = np.array([[1., 2.], [3., 4.]])
print(a.shape, b.shape)
```

→ (3,) (2, 2)

Numpy Array

- Create some commonly matrix types

```
a = np.zeros([2, 2])      # Array with 2 rows and 2 columns, filled with zeros
b = np.ones([3, 2])       # Array with 2 rows and 2 columns, filled with ones
c = np.full([4, 3], 9)    # Array with 4 rows and 3 columns, filled with 9
d = np.eye(5)                          # Identity matrix
e = np.random.random([3, 3])           # Random 0..1
```

Indexing

- Access list element by corresponding column (row) index:

  a[1, 2] = 5

- Can use index -1, -2, … like Python list:

  a[-1] = [1, 2, 3]

- Slicing column/row:

  a[:, 1:5] = [1, 2, 3, 4]

  a[3:4, 1:3] = [0, 0]

Indexing

- Select elements by index list (pay attention to dimensional index) a[<row index list>, <column index list>]
- Example:

  a = np.array([[1, 2], [3, 4]])

  a[[0, 0], [0, 1]] = 5 # a[0, 0] = a[1, 0] = 5

  → a = [[5, 2], [5, 4]]

Indexing

- Use mask to select elements

```
a = np.array= [[1, 2, 3], [4, 5, 6]]
b = a > 5
→ b = [[False, False, False], [False, False, True]]
a[b] = 0                    # a[a>5] = 0
→ a = [[1, 2, 3], [4, 5, 0]]
```

Mask is a matrix with the same shape of a, the element has a value of True/False depending on whether the corresponding position is satisfied or not.

Basic operations

- +, -, *, / → element-wise operation
- Dot product (pay attention to shape)

  np.dot(a, b)

- sum, max, min, mean (parameter axis)

  np.sum(a, axis=0)

  np.sum(b)

Basic operations

- Transpose matrix: a.T
- Repeat matrix: tile

```
a = np.array([[1, 2], [4, 5]])

np.tile(a, 2)
```

→ a = [[1, 2, 1, 2], [4, 5, 4, 5]]